

Continuum multiscale model of root water and nutrient uptake from soil with explicit consideration of the 3D root architecture and the rhizosphere gradients

Trung Hieu Mai  · Andrea Schnepf ·
Harry Vereecken · Jan Vanderborght

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Abstract

Background and aims Although modelling of water and nutrient uptake by root systems has advanced considerably in recent years, steep local gradients of nutrient concentration near the root-soil interface in the rhizosphere are still a central challenge for accurate simulation of water and nutrient uptake at the root system scale. Conventionally, mesh refinement is used to resolve these gradients. However, it results in excessive computational costs. The object of the study is to present a multiscale approach which resolves the steep gradient of nutrient concentrations at rhizosphere scale and simulates nutrient and water fluxes within the entire root zone at macroscale scale in a computationally efficient way.

Methods We developed a 3D water and nutrient transport model of the root-soil system with explicit consideration of the 3D root architecture. To capture the nutrient gradients at root surfaces, 1D axisymmetric soil models at rhizosphere scale were constructed

and coupled to the coarse 3D root-system-scale simulations using a mass conservative approach. The multiscale model was investigated under different scenarios for water and potassium (K^+) uptake of a single root, multiple roots, and whole 3D architecture of a *Zea mays L.* root system in conditions of dynamic soil water and different soil buffer capacity of K^+ .

Results The steep gradients of K^+ concentrations were efficiently resolved in the multiscale simulations thanks to the 1D model at the rhizosphere scale. In comparison with the refinement method, the multiscale model achieved a significant accuracy of K^+ uptake prediction with a relative error below 5%. Meanwhile, the simulation at macroscale with coarse mesh could overestimate the K^+ uptake in one order of magnitude. Moreover, the computational cost of multiscale simulations was decreased considerably by using coarse soil mesh.

Conclusions The newly developed model can describe the effect of the drying and nutrient transport in the root zone on nutrient uptake. It also allows to simulate processes in larger and complex root systems because of the considerable reduction in computational cost.

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T. H. Mai (✉) · A. Schnepf · H. Vereecken ·
J. Vanderborght
Forschungszentrum Juelich GmbH, Agrosphere (IBG-3),
D-52428 Juelich, Germany
e-mail: t.mai@fz-juelich.de

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Introduction

In 2050, annual world food production will need to increase by 60%, and 80% of this increase will need to come from higher yields (FAO 2016) at a lower input and in a sustainable manner (Hinsinger et al. 2011; Lynch 2007; Vereecken et al. 2016). Functional-structural models of the interactions between root systems and soil are increasingly used to enhance physical understanding of nutrient transport and water flow in root-soil systems, along with experiments and analysis (Dunbabin et al. 2013). Studies of nutrient transfer from soil to roots already have a long history (Barber 1962; Tinker and Nye 2000). The challenge is still to capture rhizosphere gradients in the whole root system scale simulation and to transfer results from single root scale to the larger scale efficiently and accurately. It has been known that root uptake of less mobile nutrients leads to strong gradients in nutrient concentrations in a small volume of soil around a root. This soil volume that is influenced by root activity is also called the rhizosphere (Hiltner 1904). The need for upscaling from the single root to the whole root system has long been recognised (Barber and Cushman 1981; Barber 1995; Tinker and Nye 2000).

The early single root models for nutrient uptake were developed for studying nutrient acquisition (Barber 1962, 1981; Passioura 1963). The nutrient transport and uptake were simulated with 1-dimensional radially symmetric models that simultaneously considered convection and diffusion in the soil as well as active nutrient uptake at the root-soil interface according to Michaelis-Menten kinetics. Although these models can be solved numerically (Passioura and Frere 1967), approximate analytical solutions were also derived along with upscaling techniques that link them to the scale of the root system (Roose et al. 2001, 2004, 2009). Recent numerical models have considered multi-component reactive transport around single roots to account for instance for the effect of root exudation on the nutrient uptake (Nowack et al. 2006; Szegedi 2009; Schnepf et al. 2012). However, in most single root nutrient uptake models, the water content and the water flow towards the root-soil interface are assumed to be constant over time. Consequently, dynamics of the water flow and the soil water content cannot be considered, although the importance of water dynamics on nutrient transfer in soil has been

recognised. For relatively mobile nutrients, redistribution of nutrients is mainly driven by water flow in the root zone. But also for less mobile nutrients, the soil water content is a critical factor since it determines the effective diffusion coefficient of these nutrients in the soil. Studying the root water uptake, Gardner (1960) presented an analytical solution of single root model for water flow towards the root surface and explained the effect of the soil properties on transpiration rate. It is also known that root biological processes can alter rhizosphere hydraulic properties, such as the exudation of mucilage (Carminati et al. 2016), or intensified aggregation through increased microbial activity (Helliwell et al. 2014). Although several nutrient uptake models for single roots have been developed and tested, it is still not practical to apply them to problems at the whole root system (macroscopic) scale due to the difficulty in describing the root architecture and interactions with soil that are relevant at larger scales.

At the root system scale, water uptake has been predicted by using root density profiles in the soil to derive sink terms in the Richards equation with the assumption of a uniform root density in a given soil control volume (Feddes et al. 1976; Clausnitzer and Hopmans 1994). Lately, the effect of root architecture, i.e., the spatial distribution of roots in soil, on resource acquisition was recognised since the local competition between roots may have an impact on the spatial pattern of water and nutrients in the soil (Baldwin et al. 1973; Javaux et al. 2013). Also, the heterogeneous distribution of nutrient and soil physical properties influence the root architecture development (Passioura 2002). For instance, roots have been shown to have a higher branching density in patches with higher nutrient concentration or to show increased growth inside bio-pores with less soil penetration resistance (Kautz et al. 2013). In the last decades, the active development of 3D root architecture modelling has provided numerous tools for studying root-soil interactions (Diggle 1988; Leitner et al. 2010a; Postma et al. 2017; Schnepf et al. 2018). The most common approach is to represent the root system as a network of small root segments. Based on the work of Doussan et al. (2006) with a simulation of water flow in a 3D root system, nowadays, the resource acquisition model has been advanced and can provide more details on the uptake processes by coupling a root architecture with a 3D soil water flow and solute transport model (Javaux

et al. 2008). Although the water and nutrient transport processes and the effect of the root architecture on these processes can be represented, the spatial resolution is typically in the order of cm or larger such that small rhizosphere gradients in the order of mm cannot be resolved. Consequently, the dynamics of water flow and nutrient gradients in the rhizospheres cannot be captured due to the limitations of resolution (de Willigen et al. 2018). Grid refinement could help to overcome this issue (Schröder et al. 2009). However, the simulation for large root system may become very computationally intensive.

The primary goal of this paper is to present a continuum multiscale model that explicitly considers the 3-dimensional root architecture, water, and nutrient flow in both soil and roots as well as rhizosphere gradients around each root segment in a computationally efficient way. Exploiting single root models for each root segment, the multiscale model can represent not only water flow and nutrient transport at the root system scale (macroscopic scale) but also the steep nutrient gradients and dynamics in the rhizospheres. In the first part, we present the fundamental equations for the continuum multiscale model. Then, the coupling method between the two considered scales will be explained. In the second part, we set up numerical experiments to study the performance of the multiscale model for roots with different complexity: from a single root to a cluster of roots and a full 3D root architecture. Nutrient uptake is also simulated for various soil conditions: from static water contents to drying soil driven by root water uptake and this for different nutrient mobilities. Here we focus on the effect of transient soil moisture content and water flux on nutrient uptake and the computational advantages of using the multiscale modelling approach.

Material and methods

Mathematical models of root-soil interactions

In this section, we present the concepts and mathematical models for water and nutrient flow at the root system scale as well as at the scale of the rhizosphere around single root segments. Furthermore, the coupling approach between the two scales is described.

Root-soil interaction model at the root system scale (macroscopic scale)

The root-soil interaction model describes the physical and biochemical processes in the soil-root system at the macroscopic scale. The model describes how these processes react (i) to boundary conditions such as the infiltration and evaporation of water at the soil surface, the transpiration of water from the canopy, the application of nutrients at the soil surface and (ii) to internal changes of the system that are related to changes of the root system due to growth, development and senescence. In this paper, we focus on water flow and nutrient transport processes in both root and soil whereas the adaptation or physiological changes of the root system to spatial and temporal variations in water contents and nutrient concentrations in the soil is out of the scope of this paper.

Equation describing water flow in soil Soil is a porous medium which consists of three phases: solid phase, liquid phase, and air phase. In this study, we consider that the soil solution is incompressible and comprised of water and a nutrient component which has a small mass fraction compared to water. Thus, we assume that the density of soil solution remains constant and equal to the density of pure water. The soil water flux, $\mathbf{u}$ [L T^{-1}], at the macroscale is driven by pressure gradients and gravitational force and can be described by the extended Darcy law

$$\mathbf{u} = -\frac{\kappa_r \kappa}{\mu} (\nabla p_{sw} - \rho \mathbf{g}) \quad (1)$$

where κ_r [-] is the relative permeability, κ [L^2] is the intrinsic permeability tensor of the soil matrix, μ [$\text{M T}^{-1} \text{L}^{-1}$] is the dynamic viscosity, p_{sw} [$\text{M L}^{-1} \text{T}^{-2}$] is the soil water pressure, ρ [M L^{-3}] is the water density and $\mathbf{g}$ [L T^{-2}] is the gravitational acceleration vector. Conservation of mass leads to the Richards equation

$$\phi \frac{\partial S_w}{\partial t} + \nabla \cdot \mathbf{u} = Q_{root-soil} \quad (2)$$

where ϕ [-] is the soil porosity, S_w [-] is the water saturation, $Q_{root-soil}$ [T^{-1}] is the source/sink term

in the soil that represents water release/uptake by the roots to/from the soil ($Q_{root-soil}$ is positive for water release and negative for uptake by the plant roots). To solve the Richards equation, constitutive relations are needed. Firstly, the water pressure p_{sw} and water saturation S_w have a relationship via a link to capillary pressure p_c [$M L^{-1} T^{-2}$] which is defined as the difference of soil air pressure p_a and water pressure $p_c = p_a - p_{sw}$. In this work, we do not consider the movement of soil air and assume that soil air pressure is constant and has the value of standard atmosphere 10^5 Pa. The relationship between capillary pressure and water saturation is described by van Genuchten (1980). Secondly, the relative permeability is linked to the water saturation by the Mualem (1976) model.

Equation describing water flow inside the root system

In modelling water and nutrient transport in the root-soil system, the diameters of root segments are usually small compared to the size of the soil control volumes. Therefore, we assume that the root system can be represented as a one-dimensional network which is embedded in the 3D soil domain. To describe the flux distribution in a root system, we stitch together multiple 1-dimensional root segments. The root is assumed to be rigid. Therefore, we neglect the effect of water pressure on the root volume. The water flow within a single root segment is described by Doussan et al. (2006)

$$-\frac{\partial}{\partial \zeta} \left[K_x \left(\frac{\partial p_{rw}}{\partial \zeta} + \rho g \frac{\partial z}{\partial \zeta} \right) \right] = q_{soil-root} \quad (3)$$

where K_x [$M^{-1} L^2 T$] or [$m^4 Pa^{-1} s^{-1}$] is the axial hydraulic conductance of the root xylem, p_{rw} [$M L^{-1} T^{-2}$] is the root water pressure, $q_{soil-root}$ [$L^2 T^{-1}$] is the source/sink term in the root that represents the water flow from/to the soil (positive for water taken up by the root and negative for water released by the root). The ζ is the arc length of the centreline of a single root segment with $\zeta = 0$ at the distal end of each root segment. The boundary conditions for the root system are set as Neumann conditions which imply zero fluxes at root tips and transpiration rate at the root collar. Water flow at all node is computed by applying Kirchhoff's law stating that the water volume flowing into each node is equal to the water volume flowing out of the node.

Equation describing nutrient transport in soil The nutrient transport in soil is described by the convection - dispersion equation:

$$\frac{\partial [(\phi S_w + b) c]}{\partial t} - \nabla \cdot (D_{eff} \nabla c) + \nabla \cdot (c \mathbf{u}) = Q_{c\ root-soil} \quad (4)$$

where c [$M L^{-3}$] is the mass concentration of nutrient, b [-] is the soil buffer power describing linear sorption isotherm process of nutrient to the soil solid phase (Barber 1995), $Q_{c\ root-soil}$ [$M L^{-3} T^{-1}$] is the nutrient source/sink term in the soil that represents nutrient release/uptake by the roots to/from the soil (positive for a source or nutrient release to the soil and negative for a sink or nutrient uptake from the soil). The effective dispersion coefficient D_{eff} [$L^2 T^{-1}$] is the combinational effect from the molecular diffusion D_m of the nutrient in soil and the hydrodynamic dispersion $\mathbf{D}_h$. The molecular diffusion D_m is affected by the tortuosity and saturation state in the soil matrix and derived by Millington and Quirk (1961) approach, $D_m = \phi S_w \tau D$, where D [$L^2 T^{-1}$] is the diffusion coefficient of the nutrient component in pure water and $\tau = \frac{1}{\phi^2} (\phi S_w)^{\frac{7}{3}}$ [-] is tortuosity of the soil matrix. The hydrodynamic dispersion $\mathbf{D}_h$ is affected by the soil pore structure and the water volume flux. $\mathbf{D}_h$ is a second rank tensor which has entries $D_{hij} = [\lambda_T \delta_{ij} + (\lambda_L - \lambda_T) \frac{u_i u_j}{|\mathbf{u}|^2}] |\mathbf{u}|$, where λ_L [L] and λ_T [L] are the longitudinal and transverse dispersivity length, δ_{ij} is the Kronecker delta, u_i denotes the i^{th} component of the water flux vector in 3D $\mathbf{u}$ (Bear and Cheng 2010).

Interactions between roots and soil: Water and nutrient uptake The interactions between roots and soil are described through the source and sink terms. The water uptake per unit root length is given by:

$$q_{soil-root} = 2\pi r K_r (p_{sw} - p_{rw}) \quad (5)$$

where K_r [$M^{-1} L^2 T$] or [$m Pa^{-1} s^{-1}$] is the radial hydraulic conductivity. In the nutrient acquisition process, we assume that passive uptake is negligible and active uptake takes the main role. Thus, we describe the source term for nutrient uptake per unit root length by the Michaelis Menten kinetics given by:

$$q_{c\ soil-root} = 2\pi r \frac{V_m c_{int}}{K_m + c_{int}} \quad (6)$$

where c_{int} [$M L^{-3}$] is the nutrient concentration at root-soil interface, V_m [$M L^{-2} T^{-1}$] is the maximum

nutrient uptake rate, K_m [M L⁻³] is the Michaelis Menten constant, $q_{c\ soil-root}$ [M L⁻¹ T⁻¹] is the nutrient uptake per unit root length. Several root segments may be located inside a given soil control volume. Therefore, the source/sink terms for water and nutrient uptake from soil are given by

$$Q_{root-soil} = -\frac{1}{V_s} \sum_{i=1}^n (l_i q_{soil-root}) \quad (7)$$

$$Q_{c\ root-soil} = -\frac{1}{V_s} \sum_{i=1}^n (l_i q_{c\ soil-root}) \quad (8)$$

where n [-] is the number of segments in the soil control volume V_s [L³], l_i [L] is the length of the i^{th} root segment.

For the water flow, the flow equations in the root system (3) and soil (2) are coupled in two ways since the source/sink terms depend on both water potentials in the soil and in the root. For the nutrient transport equations, sink source/sink terms depend only on the concentrations at the soil-root interface but not on the concentrations in the root or the transport in the root system. For both the flow and the transport equations, the water potentials and nutrient concentrations at the root-soil interface are needed for calculating the source/sink terms. When the gradients of water potential or nutrient concentration are not too large, they can be approximated by the average water potential or nutrient concentration in the soil control volume.

Water and nutrient transport around a single root segment: single root model

For less mobile nutrients, steep gradients of nutrient concentration may develop at the root-soil interface. Those are defined as the extension of the rhizosphere around each root segments. They can only be captured by the model if the computational soil domain is discretised with very high resolution, especially at locations close to the root-soil interface. As a result, the computational time and memory requirements can increase dramatically. To achieve a better water and nutrient uptake prediction with higher accuracy at low computational cost, we developed a multiscale coupling approach, in which we resolve the steep nutrient gradients using 1-dimensional radially symmetric single root models for each root segment. Then, the nutrient concentration values and source/sink terms at all of the root surfaces are fed back to the macroscale

model for the calculation of the uptake sink term. In 1-dimensional radial coordinates, water flow and nutrient transport towards a root segment are given by

$$\phi \frac{\partial S_w}{\partial t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r \kappa_r \frac{\partial p_w}{\partial r} \right) = 0 \quad (9)$$

$$\begin{aligned} \frac{\partial [(\phi S_w + b) c]}{\partial t} - \frac{1}{r} \frac{\partial}{\partial r} \left(D_{eff} r \frac{\partial c}{\partial r} \right) \\ - \frac{1}{r} \frac{\partial}{\partial r} \left(c r \kappa_r \frac{\partial p_w}{\partial r} \right) = 0 \end{aligned} \quad (10)$$

where r [L] is the radial coordinate with origin at the centre of the root segment. This model is solved in each time step for each root segment with the concentration gradient of the previous time step as the initial condition. To obtain a mass conservation between macroscopic scale and single root scale models, we transfer macroscopic fluxes at soil voxel faces to the outer boundaries of the soil cylinders by applying Neumann conditions at outer boundaries. At inner boundary (at root-soil interface), r_0 , the water and nutrient uptake are prescribed according to the following equations

$$\kappa_r \frac{\partial p_w}{\partial r} = K_r (p_{rw} - p_{sw}) \text{ at } r = r_0 \quad (11)$$

$$D_{eff} \frac{\partial c}{\partial r} + c \kappa_r \frac{\partial p_w}{\partial r} = \frac{V_{max} c}{K_m + c} \text{ at } r = r_0 \quad (12)$$

At the outer boundary, R , water and nutrient fluxes are prescribed according to the local macroscopic fluxes as follows: First, the value of the outer radius R_i of the soil cylinder around the i^{th} root segment is set by distributing the volume of the soil control volume V_s across all the root segments inside it proportionally to their radii such that

$$R_i = \sqrt{\frac{r_i^2}{\sum_{j=1}^n l_j} \frac{V_s}{\pi}} \quad (13)$$

In case of identical root radii in the soil control volume, the radii of all the soil cylinders are also identical and equal to $R = \sqrt{\frac{V_s}{\pi \sum_{i=1}^n l_i}}$ which is known as the half-mean distance between roots. Secondly, we perform a mass conservative coupling approach such that the total fluxes at the outer boundaries of the soil cylinders must be equal the net flux into the macroscopic soil control volume

$$F_{sw} = \sum_{i=1}^n F_{i_w} \quad (14)$$

$$F_{sc} = \sum_{i=1}^n F_{i_c} \quad (15)$$

where F_{sw} , F_{sc} [M T^{-1}] are the net fluxes of both water and nutrient into the soil control volume, i.e. the divergence of the fluxes in the soil control volume, F_{iw} , F_{ic} [M T^{-1}] is the fluxes of water and nutrient at the outer boundary of the i th soil cylinder. We assume that the total flux into the soil control volume is redistributed proportionally to the surface area of the soil cylinder around the roots, i.e.,

$$F_{iw} = w_i F_{sw}; F_{ic} = w_i F_{sc} \text{ at } r = R_i \quad (16)$$

where $w_i = \frac{R_i l_i}{\sum_{j=1}^n R_j l_j} \left(= \frac{r_i l_i}{\sum_{j=1}^n r_j l_j} \right)$ [-] is the weighting factor of surface area of soil cylinders or root surface. The boundary conditions at outer boundary of soil cylinders are

$$\kappa_r \frac{\partial p_w}{\partial r} = F_{iw} \text{ at } r = R_i \quad (17)$$

$$D_{eff} \frac{\partial c}{\partial r} + c \kappa_r \frac{\partial p_w}{\partial r} = F_{ic} \text{ at } r = R_i \quad (18)$$

In case of more than one root in a soil control element, the volume of the soil control volume is divided such that each root has a surrounding soil cylinder. We made assumptions that the roots are evenly distributed and that the effect of gravity can be neglected. The first assumption leads to an overestimation of uptake when roots are in reality clustered inside the soil control volume. However, we could show that this has a negligible effect on the overall cumulative uptake for the root system we considered. The second assumption is valid for high water uptake rate of a small root segment. It comes down to assuming an axisymmetrical flow field around the root segment which can be justified by the magnitude of the radial hydraulic gradients around the roots which are on the small length scale of the voxel considerably larger than the gravitational head differences (Schröder et al. 2009; Roose and Fowler 2004). It is worth to notice that these simplifications are only done inside one soil control volume. On the root system scale, gravity effect is considered in Eqs. 2 and 4.

Numerical implementation

Both the macroscopic model, given by Eqs. 2, 3, 4, and the 1-dimensional radially symmetric model, given by Eqs. 9 and 10, were solved simultaneously using the open-source DuMu^x simulation framework (Flemisch et al. 2011; Koch et al. 2018) to compute the water

flow and nutrient transport in soil and root system. DuMu^x has been developed as a module of the partial differential equations solver DUNE (Bastian et al. 2006) for multiphase flow in porous media using finite volume methods. On the macroscale, the 3-dimensional soil domain was discretised using a structured grid with hexahedral elements. The root system was represented as a one-dimensional network in a 3D soil domain and generated by CRootBox (Schneppf et al. 2018). The water flow and nutrient transport equations were solved implicitly using the cell-centred finite volume method which holds the mass conservation in each control volume of soil and root.

In the multiscale coupling approach, the macroscale model starts the simulation with 3D soil grid domain (Fig. 1a). The root system is subdivided into root segments equal or smaller than 1 cm length. For the whole root system scale, this serves as a numerical grid on which xylem flow is computed. Every root segment in a given soil control volume has a single root model assigned to it which describes 1D radially symmetric water flow and solute transport through the soil towards the root surface (Fig. 1b). The radius of the cylinder is calculated based on the soil control volume and number and geometries of other roots which together exploit water and nutrients in the soil volume. The soil radial model is performed to solve water and nutrient transport equations, capture the nutrient gradient in the rhizosphere and derive the uptake rate for the root system scale model. At the beginning of each time step, the net macroscale water and nutrient fluxes into each soil control volume (F_i) are redistributed to the cylinder mantles of the soil cylinders surrounding each single root segment inside this soil control volume according to Eq. 16. Then, the single root model is called for each root segment to compute the water potentials and nutrient concentrations at each root-soil interface. That information is fed back to the macroscale model for computing the source/sink terms Q_i (7) and (8) in the soil and root models at the next time step (Fig. 1c). In this way, our coupling method guarantees mass conservation in the soil-root system.

Numerical experiments

In order to test the multiscale continuum coupling method, a set of scenarios was developed. We considered potassium (K^+) uptake by a mature maize

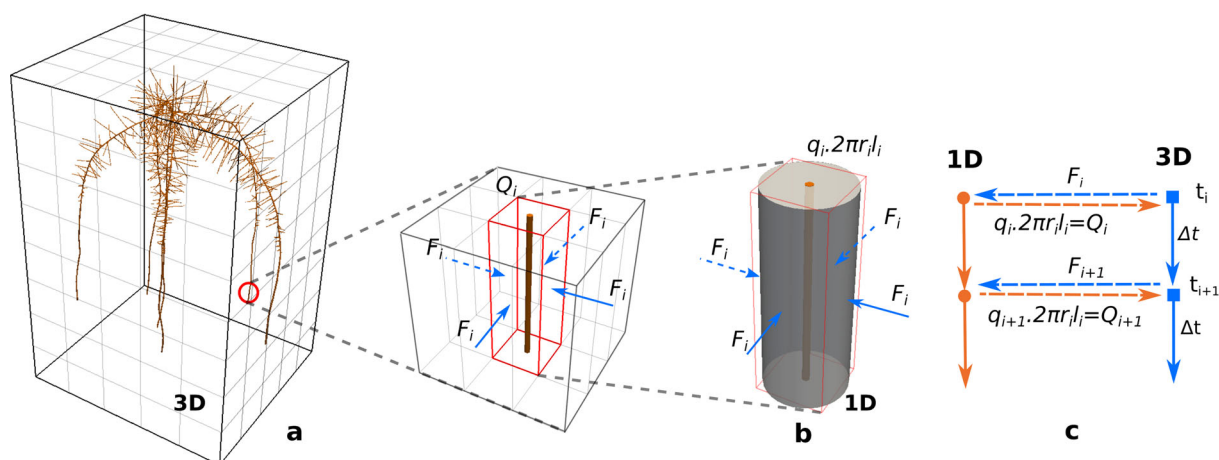


Fig. 1 The schematics of the multiscale modelling approach showing nutrient fluxes in a regular 3D grid of the macroscopic scale model (a) coupled with a 1D single root model (b), equivalent volume to the innermost soil control element. The nutrient fluxes F_i into soil volumes at macroscale are transferred to the

1D radially symmetric single root models as boundary fluxes at the outer radii. The uptake rate q_i from the single root models is transferred to the sink term Q_i in the macroscale model (c). The coupling values between two scales are exchanged within the same time step level in the simulation

(*Zea mays* L.) root system grown in a loamy soil. The plant uptake parameters for K^+ , initial K^+ concentration $c_{initial}$, buffer b and diffusion coefficient D were taken from measurements of field grown maize (Barber 1995). The hydraulic conductivity of maize was taken from French and Steudle (1989). The maize roots were considered to be non-growing during the simulation period of 20 days. Numerical experiments for evaluation were performed for a single root, for multiple parallel root segments, and finally, for a complex 3-dimensional root system. To demonstrate the features of the coupling method, the nutrient uptake from the multiscale model were compared with simulations that use only the macroscale model and simulations using only the single root scale model. Different grid sizes were considered to evaluate the effect of spatial discretization on the accuracy of the simulation results from the macroscale model. In simulations that use only the single root scale model, no-flux boundary conditions were applied at the outer boundaries of the soil cylinders, i.e., macroscopic water flow and solute transport are ignored.

Scenario 1: Water and nutrient uptake by a single root

To evaluate the multiscale coupling approach for nutrient uptake by a single root, we constructed a scenario for which we could use the classical model of Barber and Cushman (1981) as a reference. We considered a

root with 3 cm length and 0.02 cm radius and located at the centre of a soil cube sized $3 \times 3 \times 3 \text{ cm}^3$. We considered homogeneous initial conditions and Dirichlet boundary conditions at the faces of the soil cube. The 3-dimensional simulation using macroscale approach was run with different soil discretization sizes Δx : 1 cm, 0.5 cm, 0.1 cm. On the other hand, the simulation of the multiscale continuum model was performed with a coarse soil discretisation sizes of 1 cm. In each single root model, the radial distance between root surface and outer boundary was discretized by 10 elements using a logarithmic scale with grading factor 1.5. The discretisation size is reduced 1.5 times in each element towards the root surface to get the highest resolution at the root-soil interface. The nutrient uptake was calculated for different soil conditions: static soil water contents θ : 0.4, 0.2, 0.1 and drying soil caused by a constant root water uptake rate of $10^{-5} \left[\frac{\text{cm}^3}{\text{s cm}^2} \right]$. The mobility of nutrient in soil was also varied by considering different soil buffer powers b : 20, 40, 80. All simulation results were evaluated with respect to the reference model. The reference model was run with soil discretisation of 0.003 cm.

Scenario 2: Water and nutrient uptake by multiple roots in a quasi-2D domain

To evaluate the multiscale coupling approach in the presence of multiple roots within one soil control

volume of the macroscale model, we constructed a scenario for which we could use an explicit 2-dimensional simulation as a reference. We first generated a maize root system using the root system model CRootBox (Schnepf et al. 2018). Then, we chose one horizontal cross-section through the simulated root system and extracted the position and radii, which varied from 0.04 to 0.1 cm, of 64 roots that crossed that cross-section. From this information, we constructed the reference simulation (also in DuMu^x) in a quasi-2D soil domain-sized $36 \times 40 \times 1 \text{ cm}^3$, i.e., with only 1 cm in the z-direction and assuming that the roots are vertical in this domain. In the reference simulations, the geometry of the boundaries between the soil and roots is explicitly considered, and boundary conditions for water flow and nutrient transport are defined at the soil-root interface to solve the flow and transport equations in the soil domain. An unstructured mesh with 24847 elements was constructed for this simulation. The soil mesh was refined near the root surfaces with element size one order of magnitude smaller than the radii to capture the nutrient gradient at the root surface. To set up the multiscale continuum coupling approach, we used the same root clustering pattern and soil domain as in the reference simulation but considered a regular grid for the macroscopic model. The macroscopic model was set up with different soil discretization sizes $\Delta x = \Delta y : 4 \text{ cm}, 2 \text{ cm}, 1 \text{ cm}$ and 0.5 cm . On the other hand, the soil mesh in the multiscale coupling model was discretised with the coarsest spatial step of 4 cm. The spatial domains of the cylindrical soil model were defined by Eq. 13 which depends on the length and radius of roots in the soil voxel. The soil cylinder domains were discretised by ten elements using a logarithmic scale with grading factor 1.5. The spatial resolution close to the root surface was about 0.05 cm. The soil had an initial water content, θ_{initial} , of 0.3 [-]. We assumed a constant water flux across the root-soil interface of $7.27 \cdot 10^{-6} \left[\frac{\text{cm}^3}{\text{s cm}^2} \right]$ which corresponds to the water uptake rate $1.6 \cdot 10^{-4} \left[\frac{\text{cm}^3}{\text{s}} \right]$ of the root clusters and $5 \cdot 10^{-3} \left[\frac{\text{cm}^3}{\text{s}} \right]$ of the whole root system. The water and nutrient in the soil were assumed to be limited. Thus, zero flux boundary conditions were implemented at all faces of the soil domain.

To demonstrate the effect of nutrient transfer at the macroscopic domain, we set up a scenario of nutrient uptake in a soil with a low buffer capacity ($b=5$)

and a static water content with zero transpiration rate of roots. Thus, the nutrient transfer toward roots is driven only by diffusion. We compared the multiscale approach with the one using only the single root models for each root segment to test the error of ignoring macroscopic flow under conditions of mobile nutrients.

Scenario 3: Water and nutrient uptake by the full root system in 3D

The full 3D maize root architecture was generated using root architecture parameters from Leitner et al. (2014). The maize root system had a total length of 2200 cm, consisted of 2208 root segment length in a range of 0.01–1 cm. Some root segments are smaller than the user-defined standard root segment length (here: 1 cm) in order to accommodate new branches in specific positions. The root system was located in a $36 \times 40 \times 60 \text{ cm}^3$ soil domain. The root radii varied from 0.03 to 0.14 cm. The soil was considered to be homogenous in hydraulic properties and buffer power for K^+ $b=40$ [-]. The initial K^+ concentration in soil c_{initial} was $0.5 \left[\frac{\mu\text{mol}}{\text{cm}^3} \right]$. The nutrient uptake was investigated in the condition of a drying soil by plant transpiration. The transpiration rate was applied at the root collar, and the soil was assumed to be limited in water and nutrient by applying zero flux at its boundaries. All model input parameters are given in Table 1.

Results

Evaluation of the multiscale coupling approach for a single root

Water and K^+ uptake by a single root in drying soil

Figure 2 shows the relative K^+ concentrations with respect to distance from the root surface as simulated by the macroscale model, the multiscale model and the Barber-Cushman model as a reference (left panel) in case of soil drying due to transpiration. The soil water content decreased from 0.4 to 0.14 after 20 days. The right panel shows the Darcy water flow velocities in the rhizosphere caused by root water uptake after 20 days in the multiscale simulation and the reference

Table 1 Model parameters

Parameter	Value	Unit	Sources
Soil properties			
van Genuchten α	$0.04 \cdot 10^{-2}$	$[Pa^{-1}]$	Vanderborght et al. (2005)
van Genuchten n	1.6	$[-]$	Vanderborght et al. (2005)
Residual saturation S_{wr}	0.08	$[-]$	Vanderborght et al. (2005)
Intrinsic permeability κ	$5.9 \cdot 10^{-13}$	$[m^2]$	Vanderborght et al. (2005)
Porosity ϕ	0.43	$[-]$	Vanderborght et al. (2005)
Nutrient buffer capacity b	20,40,80	$[-]$	Barber (1995)
Diffusion coefficient D	$1.96 \cdot 10^{-9}$	$[m^2/s]$	
Root properties			
Axial conductance	$5 \cdot 10^{-17}$	$[m^4/Pa.s]$	French and Steudle (1989)
Radial conductivity	$2.1 \cdot 10^{-13}$	$[m/Pa.s]$	French and Steudle (1989)
Michaelis Menten kinetics			
Maximum uptake rate V_m	$3 \cdot 10^{-5}$	$[m^3/mol.s]$	Barber (1995)
Michaelis constant K_m	$14 \cdot 10^{-3}$	$[m^3/mol]$	Barber (1995)

model. Both cases capture well the increase of the absolute value of the Darcy flux towards the root surface due to the cylindrical coordinates (i.e. $2r\pi u = 2r_0\pi u_0$, where r_0 is the root radius and r is the radial distance from the root surface). Figure 2 clearly shows the strong K^+ depletion profile around the root, along with a strong increase of water flux in the rhizosphere. Using the logarithmically graded mesh, the single root model in the coupling approach successfully handled

the gradients at the root surface and captured well the depletion zone of K^+ . By contrast, the macroscale model could not capture the steep concentration gradient near the root surface. Figure 3a and b show the potassium uptake rate and the cumulative uptake from the macroscopic simulations with different spatial resolution and from the multiscale model with coarse soil discretisation but with local refinement in the single root models. The corresponding relative error with

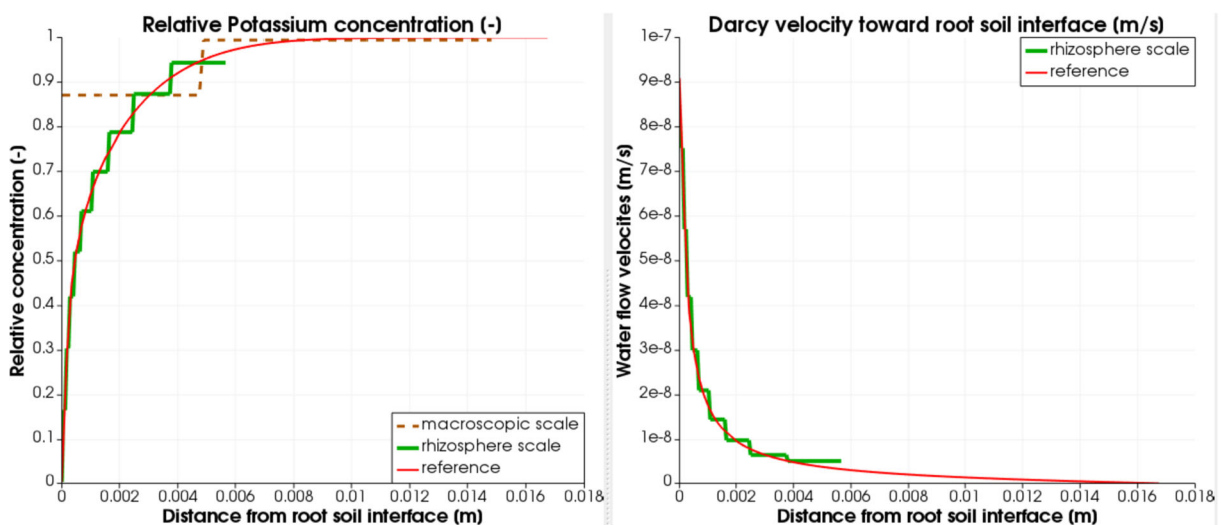


Fig. 2 Concentration profiles of potassium as computed by the macroscopic scale model, single root model within the multiscale coupling approach and the reference model (Barber

Cushman model) (left panel) at 20th day. The right panel shows the corresponding Darcy velocity in the reference model and the rhizosphere module of the multiscale coupling approach

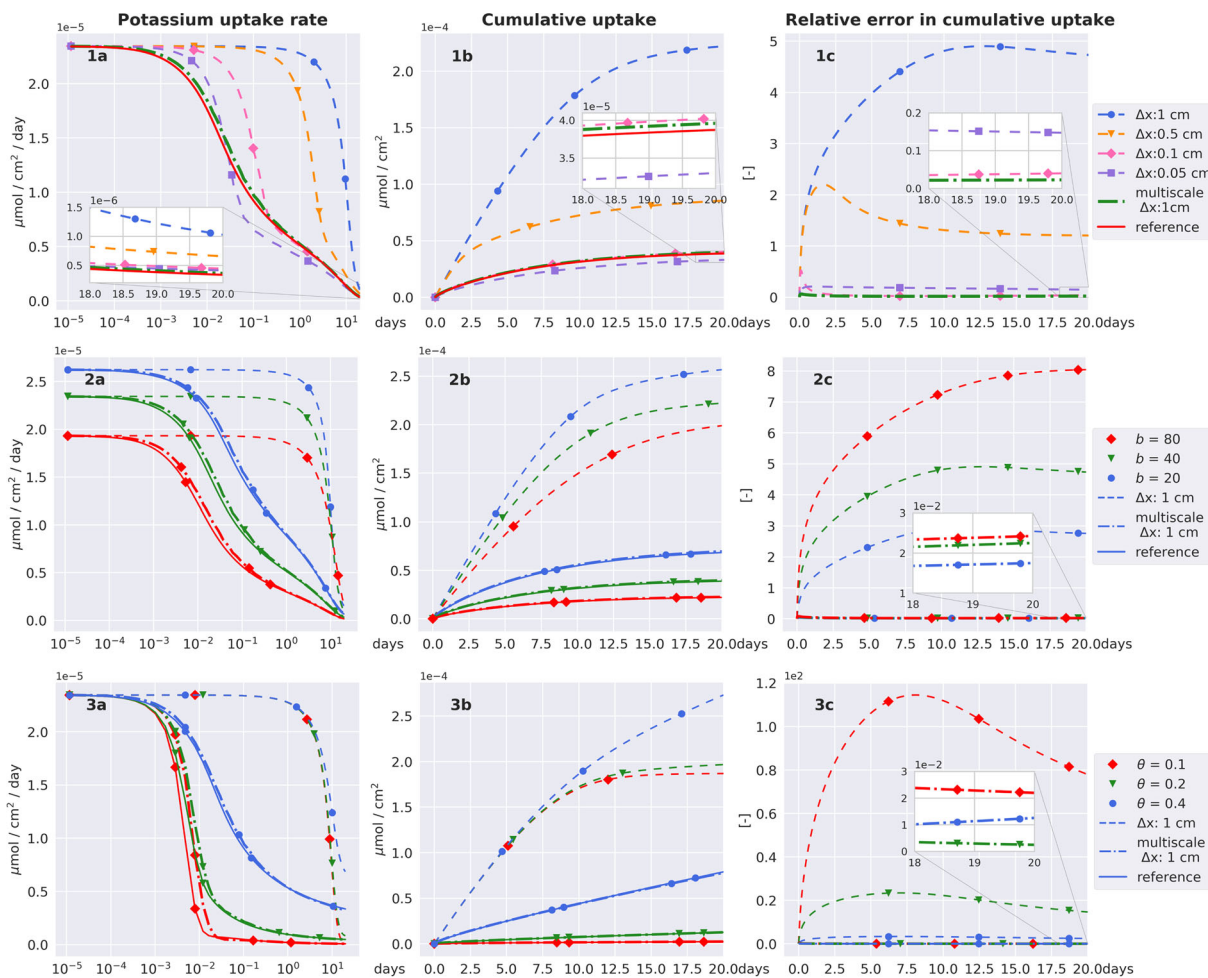


Fig. 3 The effect of different spatial discretisation Δx on the Potassium uptake rate (a) and the cumulative uptake (b) as computed by the macroscale model, the multiscale model that combines the macroscale model at 1 cm resolution with the cylindrical single root soil model for each root segment, and

the reference model. Relative errors of cumulative uptake (c) computed by the different cases compared to the result of the reference model. The nutrient uptake under different soil sorption capabilities with buffer power $b = 20, 40, 80$ (in 2nd row) and different soil water availability $\theta = 0.1, 0.2, 0.4$ (in 3rd row)

respect to the reference model is shown in Fig. 3c. The nutrient uptake rate from all simulations decreased over time. The simulation with coarse soil mesh of 1 cm overestimated the uptake rate and the cumulative uptake largely. By refining the soil mesh size to 0.1 cm, the accuracy of the macroscale simulation was enhanced significantly. However, the predicted uptake became underestimated when using the smallest soil discretisation of 0.05 cm. On the other hand, with coarse soil resolution of 1 cm, the multiscale simulation accurately predicted the nutrient uptake over time with a small relative error of 3% compared to the reference value.

The effect of varying the soil buffer power on K^+ uptake by a single root in drying soil

The effect of different soil buffer power on the nutrient uptake rate and the cumulative uptakes is showed in Fig. 3a and b. The soil resolution 1 cm was used in both of the macroscale and the multiscale model. The drying soil condition and the soil buffer power level effected strongly to the nutrient uptake. Initial condition was chosen such that the total concentration (adsorbed plus dissolved) was the same in each buffer power case. We observed that, generally, the lower the buffer power, the higher the nutrient uptake.

Regarding the accuracy, the multiscale simulation well matched with the reference values with a small relative error in the range of 3%. In contrast, the macroscopic scale model using a resolution of 1 cm highly overestimated the nutrient uptake. The most significant error was found in case of the highest buffer capacity where the relative error can reach over eight times (see Fig. 3c).

The effect of varying the water content on nutrient uptake from soil

Figure 3a and b presents the effect of different soil water content on the uptake rate and the cumulative uptake in the macroscale and the multiscale simulation. Initially, the K^+ uptake rates were the same in all simulations because of the same initial concentration. But during simulation, it can be observed that the higher the soil water content is, the more significant is the amount of nutrient that can be taken up by the plant root. The overestimation of the macroscale model can reach 1 or 2 order of magnitudes higher than the reference value (see Fig. 3c). The largest relative error can be seen clearly in case of the driest soil when the cumulative uptake was strongly limited. On the other hand, the multiscale model matched well with the reference results with minimal relative errors from 0 to 3%.

Evaluation of multiscale coupling approach for multiple roots

Figure 4 presents the nutrient depletion zones created by multiple roots in a 2D scenario from simulations of the reference model (a) and the multiscale model (b). The nutrient depletion zones developed around each root partly overlap with each other at positions with high root density. Figure 4c shows the concentration profile along the line depicted in Fig. 4a and b as computed by the multiscale model with single root scale module and the reference model. Interestingly, the depletion zones were captured quite well by the multiscale model. Figure 5 shows the uptake rate, the cumulative uptake and its relative errors from macroscopic simulations with different soil discretisations and from the multiscale model with the rhizosphere-scale module at a coarse soil mesh of 4 cm edge length. One can see a considerable overestimation of

nutrient uptake by the macroscopic model when a coarse soil resolution of 4 cm is used. The accuracy was improved by resolving soil mesh down to 0.5 cm, but the relative error was still high (about 100%). On the other hand, the multiscale simulation with 4 cm discretisation successfully captured the reference with a small relative error of 2.5%. The uptake rate is initially underestimated also by the multiscale model, but this is not significant regarding the cumulative uptake. Figure 6 presents simulation results for a nutrient of medium mobility (buffer power $b=5$). In the 2D reference (a), the depletion zones of nutrient overlapped with each other and the most depleted part locates at the centre of the root cluster. The potassium uptake rate (b) is overestimated by the macroscale model. This could be explained by the fact that the macroscopic model cannot resolve the nutrient gradients in the rhizospheres. As a result, the concentrations at the root-soil interface were not accurately approximated. On the other hand, the simulation at microscopic scale with the cylindrical soil models only (without considering macroscopic flow and transport) underestimated the uptake rate due to the lack of nutrient transfer at macroscopic scale. In contrast, the multiscale simulation gave a good prediction of nutrient uptake by resolving the nutrient gradient at root-soil interfaces.

Evaluation of multiscale coupling approach for 3D root system

Due to the complexity of the maize root system, there is no reference simulation to evaluate the accuracy of simulation results. Thus, we compared the outcomes of the multiscale model to the outcomes from the macroscopic model at different spatial resolutions. A visualisation of the multiscale simulation result is shown in Fig. 7, where one can observe the nutrient depletion zones around all the roots after 20 days on horizontal planes at 15 cm and 35 cm depth. It is worth noticing that the different spatial distributions of roots resulted in different patterns of the depletion zones. Because of the high root density at 15 cm depth, the roots compete for the nutrient resource and the depletion zones overlap each other. On the other hand, the 20-day old maize root system had fewer roots at 35 cm depth. Therefore, the roots have more soil to exploit nutrient, and the depletion zones are isolated. Figure 8 shows the uptake rate, the cumulative uptake and its

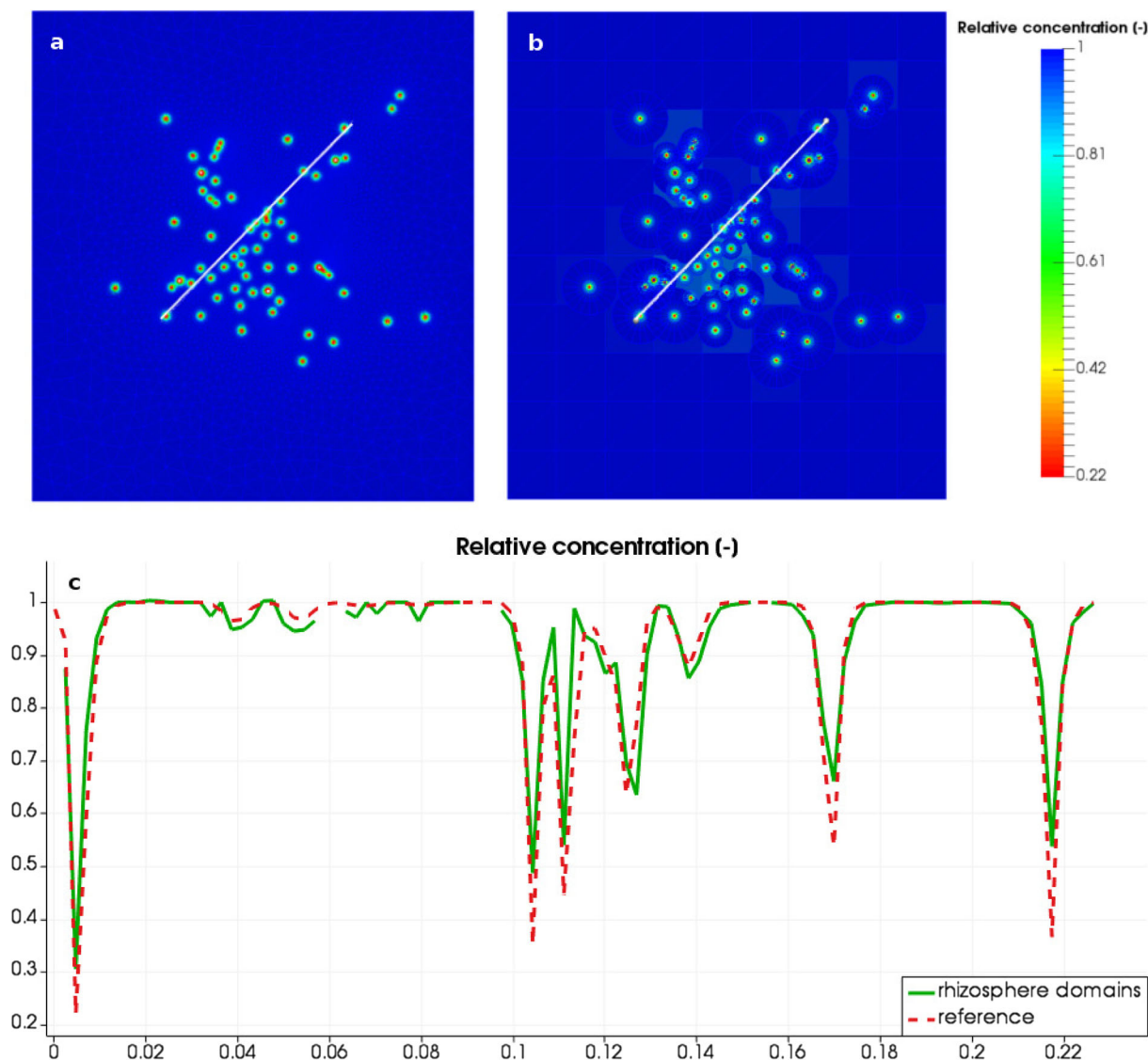


Fig. 4 Nutrient depletion zones simulated by the reference model (a), the multiscale model (b) and the relative concentration along the transect line (c)

relative difference between macroscopic simulations with different soil discretisations and the multiscale simulation with a coarse soil mesh of 4 cm. Overall, it can be recognised that using the coarsest soil mesh of 4 cm overestimated the nutrient uptake simulated by the multiscale model by one order of magnitude. By refining the soil mesh, the accuracy of predictions by the macroscopic model was improved. However, for the finest resolution of 0.5 cm, the macroscopic model overestimated the cumulative uptake simulated by the

multiscale model by a factor 2. The relative errors of the macroscopic and multiscale models in this 3D simulation are similar to the ones in the 1- and 2D simulations. Thus, we are convinced that the multiscale model also gives more accurate results in the 3D case.

In computational performance aspect, the mesh refinement increased the computational time significantly due to the exponential increase of the degrees of freedom. The macroscopic simulations with 4 cm, 2 cm and 1 cm soil resolution took 10, 45 and 340

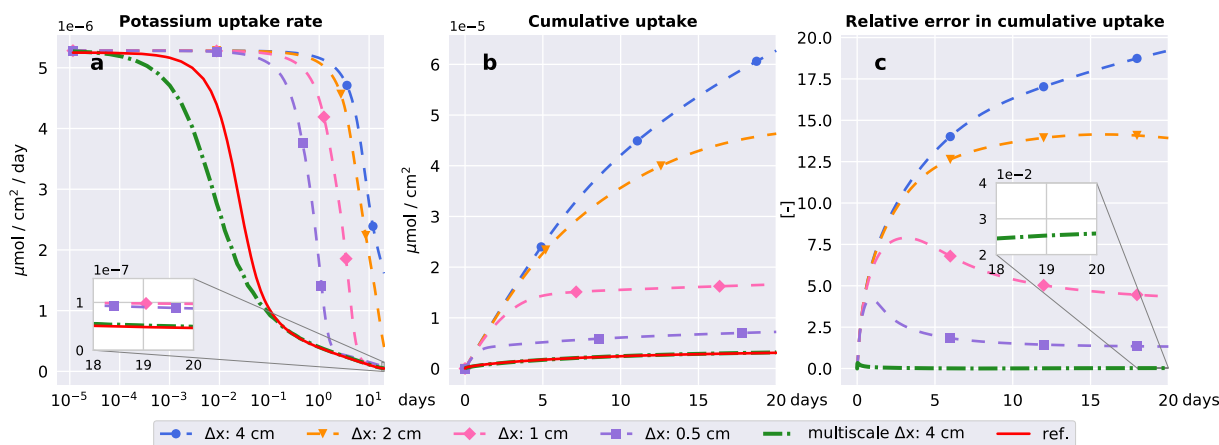


Fig. 5 The effect of different spatial discretisation on the Potassium uptake rate (**a**) and the cumulative uptake (**b**) of the cluster of root as computed by the macroscale model with refinement

approach, the multiscale model and the reference model. **c** Relative errors of cumulative uptake compared to the result of the reference model

minutes respectively. The finest discretisation size of 0.5 cm made the simulation time longer than 35 hours which implies that the refinement of the whole soil domain is infeasible for practical simulations. On the other hand, the multiscale model took only 1.67 hours for the simulation thanks to the coarsest mesh. It's also noticed that the multiscale simulation requires more computational time than the macroscopic simulation with a similar amount of degrees of freedom (10^4 elements by 2 cm discretisation). All of the simulations were performed by a computer with processor Intel Core i7-5600 (2.6GHz $\times$ 4) and 16GB RAM.

Discussion

Water and nutrient uptake for a single root

In the scenarios for the single root, we showed that the accuracy of macroscopic scale simulation strongly depends on the soil discretisation. Using a coarse soil grid of 1 cm, the cumulative nutrient uptake from macroscopic scale simulations were higher than the reference value by a factor 6–11. It is obvious that nutrient concentration at the root-soil interface in the macroscale model was overestimated due to the coarse

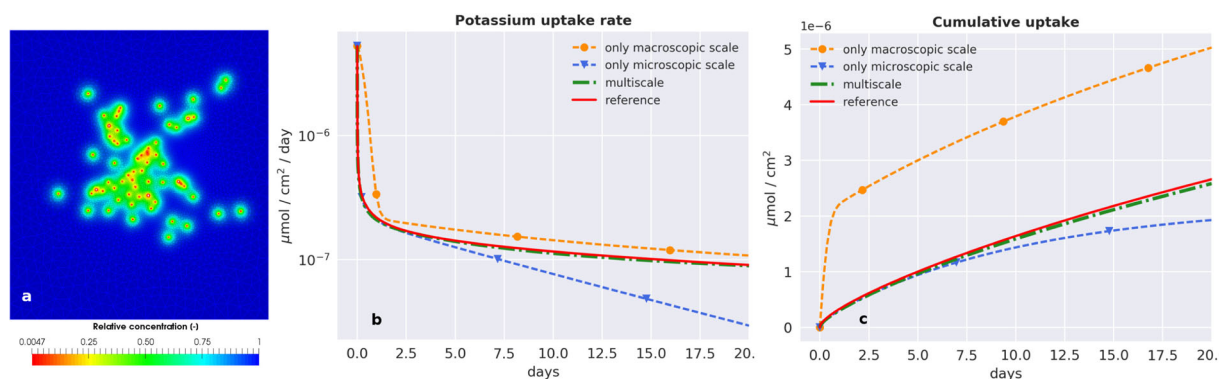


Fig. 6 The potassium uptakes from simulations with different approaches: only macroscopic model, only single root model for each root segment and the multiscale approach: (**a**) relative

concentration in soil domain (the reference value); (**b**) the potassium uptake rate over time; (**c**) the cumulative uptake over time

Fig. 7 Nutrient depletion zones within the soil domain caused by water and nutrient uptake of the maize root system after 20 days

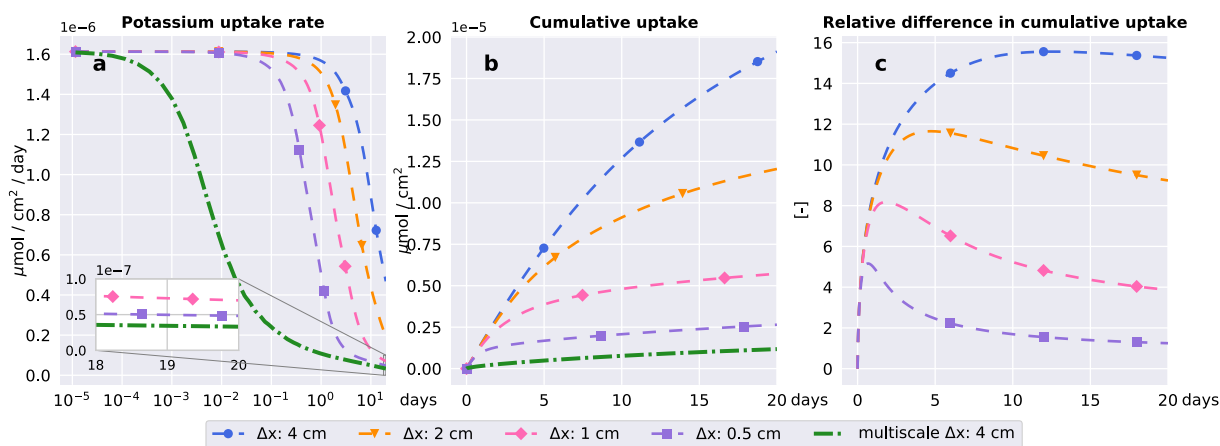
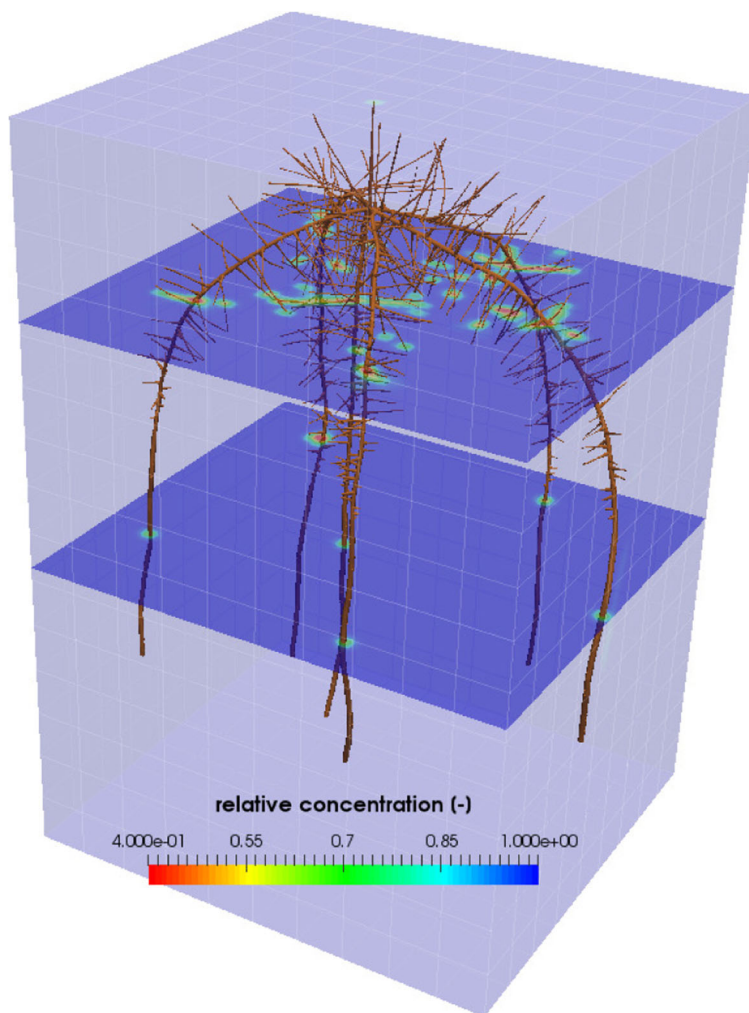


Fig. 8 Potassium uptake rate (a) and cumulative uptake (b) of the maize root system as computed by the macroscale model at different spatial resolutions and the multiscale model

(c) Relative difference of cumulative uptake computed by the macroscopic model compared to the result of the multiscale model

soil grid size. Therefore, the soil mesh needs to be refined, especially near the root surface to resolve the steep gradient of nutrient concentrations. The study on the single root scale revealed that soil mesh refinement improved the accuracy of macroscale simulation. However, when the soil grid size became comparable to the root diameter (in this case 0.5 mm), the nutrient uptake did not converge to the reference value. This can be explained by recalling the conceptual representation of the root system in the soil. In the current approach, the root volume is neglected in the soil volume. The root system are represented as a 1-dimensional network of root segments which represent line sinks in the soil. Therefore, in case of too fine soil grid size where the root volume cannot be ignored anymore, the nutrient concentration at root-soil interface can be wrongly predicted due to the lack of physical presence of the roots in the soil. That makes the refining method less reliable when the spatial resolution reaches the same order of magnitude as the diameter of individual roots. On the other hand, the multiscale approach we presented here captured well the nutrient concentration at the root-soil interface. Using the single root component, the multiscale model resolved the steep nutrient gradients in the rhizosphere, but still, exploited coarse soil element which is essential for the conceptual representation of 1D root system. The numerical experiments for single root also showed that the accuracy of nutrient uptake predictions also depends strongly on soil conditions. The highest errors were found in conditions of dry soil or high buffer capacity. In those conditions, the nutrient gradients in the rhizosphere were very steep due to the fact that the diffusion of nutrient is limited in condition of low water content or high sorption of nutrients to the soil.

Water and nutrient uptake for multiple roots

The evaluation of the 2D scenario with root clusters revealed similar overall accuracy differences between macroscopic and multiscale models, with the multiscale approach being the most accurate one. For the case when several roots are inside one soil control volume, the interaction between roots was taken into account. In soil with low buffer capacity, the nutrient becomes more mobile and the depletion zones around individual roots overlap each other. That indicates a

strong competition between roots for resources (see Fig. 6a). The simulation results have shown that the multiscale model exploited well the information of roots in the soil control volume, such as the segment lengths, the root radii, to transfer macroscopic process to the single root scale. In particular, the available soil volumes around each root segment were derived by the generalized half-mean distance between roots (13). The flow and transport from the macroscopic scale was redistributed to the rhizosphere domains based on root surface ratios between roots in the soil control volume (16). Those downscaling translations take the effect of roots competition into account and have shown its efficiency in the 2D scenarios which we had a reference solution. In the 3D scenario, we account for inter-root competition in the same way. Thus, we are convinced that the accuracy also transfers to the 3D scenario.

A similar evaluation between an explicit 2D simulation and an equivalent, upscaled approach based on cylindrical flow towards the single roots was presented for water flow by Beudez et al. (2013). They transformed the Richards equation for water flow into a linear equation and could then make use of the superposition principle. They also demonstrated the computational advantage of using a coarser mesh for the macroscopic simulation while resolving single root gradients in a 1D axisymmetric domain. In their case, they applied an analytical solution for the matric flux potential gradients around each root together with the superposition principle for linear partial differential equations to come up with the matric flux potential and corresponding water uptake by each root. The sink term is, like in our case, computed by summing over the uptake fluxes of all the roots that lie inside a given soil control volume. Their results agree in two points with ours: (a) their equivalent model in some instances overestimated uptake compared to the 2D explicit model and (b) the 2D water potential maps of the equivalent model appeared somewhat blurred compared to the explicit 2D model because it is based on average bulk values. But, the overall accuracy is rather good.

Another effort to couple root architecture nutrient uptake model with a reactive transport model was presented by Gérard et al. (2017). The study considered multicomponent chemical reactions in a root-soil interactions model. However, the sink term

for nutrient uptake was based on the average root surface area in each soil grid and did not consider small-scale rhizosphere gradients around individual roots. Our numerical study showed that those gradients are crucial for a better nutrient uptake simulation and should not be neglected. An avenue for future model development will be the combination of the two approaches.

3D functional-structural models of root architecture that have been used for computing nutrient uptake from soil include OpenSimRoot (Postma et al. 2017), ROOTMAP (Dunbabin et al. 2002) and R-SWMS (Schröder et al. 2012). Those models have in common that they use the macroscopic approach for computing root solute uptake and are thus likely to be more accurate for water and mobile ions such as nitrate. OpenSimRoot, also, offers the possibility to simulate depletion zones around a single root. However, this module is then not solved in combination with the 3D model of solute transport in the whole soil domain and is thus more likely to be suitable for strongly sorbed solutes such as phosphate. The multiscale approach in this paper represents a bridge between those two situations (macroscopic only or microscopic only) by combining macroscopic 3D water flow and solute transport with the consideration of rhizosphere gradients around individual roots. Also in the case of non-uniformly distributed roots, our new approach presents an improvement as it can simulate potentially important macroscopic nutrient flux to root clusters. Steady-state rhizosphere profiles have been considered before in the rhizosphere version of ROOTMAP (Dunbabin et al. 2006). Compared to that, our approach represents an extension because it considers the dynamic development of the rhizosphere gradients.

Our single root scenarios can be regarded as an extension of the study from Nowack et al. (2006) where different numerical solvers were compared to this approximate analytical solution. The numerical DuMu^x model we present here is more general, particularly in terms of our explicit representation of root architecture, which we create using the root architecture model CRootBox and which can also grow in response to soil conditions (not shown here, but see Schnepf et al. 2018) and in terms of the DuMu^x structure which facilitates the extension of this model e.g. to multiple components.

Discussion and opportunities for further development of the multiscale model

Our multiscale approach aims to improve the transfer of single root solute uptake to the root system scale efficiently and accurately. To verify the coupling method and for simplicity, we only considered the fundamental processes which driven the nutrient transport in the root-soil system. We are aware of the fact that we chose the "minimal single root model" (Darrah et al. 2006) and that the model itself does not yet represent all rhizosphere processes that are relevant for nutrient uptake and may have limitations to describe these processes which could be in further development.

1. Most current single root models employ continuum-based descriptions of the soil on the Darcy scale (Roose and Kirk 2009). We thus assume that solute transport in the rhizosphere can be represented by the convection-dispersion equation and water flow by the Richards equation. Our continuum approach uses effective parameters that represent upscaled pore scale processes in representative elementary volumes (REVs) at the continuum scale. Under which conditions this approach is still applicable to link average fluxes to average gradients of water potentials and nutrient concentrations when the spatial scale of changes of these average water potentials and concentrations are of the same order of magnitude as the scale of individual pores, and how alternative approaches can be formulated to describe anomalous transport is still the subject of ongoing research (Roose et al. 2016). To solve such problems, readers are referred to other approaches such as the continuous time random walk model.
2. The effect of root hairs or mycorrhizas is neglected.
3. We assumed that the root volume does not change over time and that the roots are in full contact with the soil. Thus we neglect processes such as air gap formation at the root surface when roots shrink (Carminati et al. 2013), compaction of the bulk soil around thickening roots (Aravena et al. 2011) and resulting change of hydraulic properties, and the interaction between roots and soil in larger pores of structured soils (Landl et al. 2017).

4. In our study, the buffer coefficients were assumed to be constant. However, sorption-desorption processes may be influenced by biogeochemical changes in the soil around the roots that are induced by root exudates and microbial activity. Furthermore, root exudates may also change the rhizosphere hydraulic properties with consequences for the water flow and root water uptake (Carminati et al. 2016).
5. Plant physiological processes are beyond the scope of the work. However, nutrient uptake according to Michaelis-Menten kinetics with constant coefficients is a limitation ignoring internal regulation of uptake. de Willigen et al. (2018) base root uptake on plant demand (rate of dry-matter production times the amount of required nutrients) as long as soil transport can maintain the concentration above a (very small) threshold value. This type of regulation can be straightforwardly implemented in the model by adjusting the Michaelis-Menten kinetic parameters. In the context of pesticide uptake by roots, Trapp (2000) models diffusive and convective uptake whereas diffusive uptake is dependent on concentration gradients between soil and root. Plants are also observed to regulate their uptake of nutrients in elaborate patterns according to their environment stage of development. Thus the Michaelis-Menten parameters are not constant in time (Bhattacharyya and Datta 2005). In principle, our model implementation will facilitate the implementation of such approaches. However, their parameterisation represents a challenge.

Several of the processes mentioned above could be implemented further in the single root model. Effects of root hairs could be represented by extending the root radius (Passioura 1963), or using a distributed sink term around the root (Leitner et al. 2010b). Gaps between the roots and the soil could be represented by changing the relationship between soil water content and hydraulic conductivity (Couvreur et al. 2014; de Willigen et al. 2018) and effective diffusion coefficients. Also, local compaction could be represented by changing hydraulic soil properties with distance from the root surface. The root scale model could be coupled with bio-geochemical models to simulate the

effect of root exudates on buffer coefficients (Nowack et al. 2006). Regulation of nutrient uptake by plants can be implemented by defining root segment specific and time or root segment age dependent Michaelis-Menten kinetics coefficients (Bhattacharyya and Datta 2005). Instead of using a 1-D root segment model, variations along the segment axis could be represented using a 2-D axisymmetric model whereas a full 3D model could be used when the axial symmetric conditions are not satisfied anymore. But, these changes and implementations are relatively straightforward. The multiscale modelling approach that we presented here offers a framework that can be used to evaluate the impact of these smaller scale processes on the nutrient uptake at the plant scale in a numerically efficient way. A more critical aspect is the validity of the effective soil properties (hydraulic properties and effective diffusion) and the continuum scale equations to describe processes at the sub mm scale. Continuous time random walk models could be implemented to describe these small scale processes. But this would correspond with a paradigm shift in the algorithms that are used to simulated the transport processes around single roots.

Conclusions

In this contribution, we presented a multiscale modelling approach for water flow and nutrient transport and uptake linking the single root segment scale to the whole root system scale of a maize root system, using potassium as an example. The numerical experiments showed that the dynamics of nutrient gradients at the rhizosphere scale is a great challenge in root nutrient uptake modelling. Without considering those gradients, we demonstrated that the prediction from macroscopic simulation could be inaccurate with one or higher order of magnitude. The soil conditions also showed a strong effect on the accuracy of the simulation at the root system scale. Especially in the case of drought conditions or highly immobile nutrients, the uptake at macroscale simulations had considerable inaccuracy with 1 or 2 order of magnitude higher. On the other hand, the proposed multiscale approach improved the nutrient uptake predictions significantly with a small relative error though using coarse soil discretisation at the macroscopic scale. The method

linked the transport fluxes from macroscopic scale to single root scale and resolved the gradients effectively near root surface. This study is the first and essential step for developing a model that can predict the uptake of nutrients and bridge the gap between the rhizosphere scale and macroscale in the soil-root system.

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